Error Correction and Pairs Trading Summary — Week 7 Part 3

Oct 10, 2026 · @Keehong Rhee

At a glance

Cointegration says the spread comes back; error correction explains **why** it comes back. Part 3 turns that pull back into a trading strategy and also covers the risk that the equilibrium breaks. It comes down to three points.

1. **Error correction is the restoring force.** A VECM is a differenced VAR plus a term for last period's equilibrium error. Its coefficient α (the adjustment speed) must be negative to close the gap, and its size sets the half-life.
2. **Cointegration and error correction come as a pair.** By the Granger representation theorem, cointegrated series have an error-correction representation, and the converse also holds.
3. **The spread is a trade, but it can break.** Pairs trading Stage II takes a market-neutral opposite position when the spread, standardised as a z-score, is stretched. The decisive risk is that cointegration breaks down, so re-test regularly and keep a stop-loss.

1. Starting point: from equilibrium to mechanism

Parts 1 and 2 let us decide whether a long-run equilibrium exists. Now we model **what** holds the spread in place and stops it wandering.

1. **We can confirm cointegration.** Parts 1 and 2 gave the concept and the tests. We know whether a stationary spread, that is, a long-run equilibrium, really exists.
2. **But what holds the spread in place?** When the pair strays, something has to pull it back. That is error correction, an explicit restoring force built into the dynamics.
3. **That force is the trade.** How fast and how reliably it pulls back decides whether the spread is worth trading. Pairs trading Stage II is built on exactly this force.

2. Error correction: the restoring force

When a cointegrated pair moves away from equilibrium, its next moves tend to close the gap. This built-in correction keeps the spread stationary. It is a self-correcting system.

Mechanism and intuition

- **A gap opens**: the spread moves away from 0.
- **The correction responds**: the following changes shrink the gap. The larger the gap, the stronger the pull.
- **The system remembers**: it remembers last period's deviation from equilibrium and uses that memory to steer the next change. A positive gap calls for a downward correction, a negative gap for an upward one.

- **Why it must exist**: a stationary spread cannot wander far, so some force must hold it. That force is the error-correction term. No error-correction term, no cointegration.

Two-variable error-correction model (ECM)

Each series' change is explained by "last period's gap" and short-run changes. As in the lecture figure, a gap above 0 is pulled down, and a gap below 0 is pulled up.

$$\Delta Y_t = \alpha_Y \underbrace{(Y_{t-1} - \beta X_{t-1})}_{z_{t-1}\ \text{(last gap)}} + \gamma_Y \Delta X_{t-1} + \varepsilon_{Y,t}$$

$$\Delta X_t = \alpha_X (Y_{t-1} - \beta X_{t-1}) + \gamma_X \Delta Y_{t-1} + \varepsilon_{X,t}$$

- z_{t−1} = Y_{t−1} − βX_{t−1}: last period's equilibrium error (the spread). The β estimated in Part 2 goes here.
- α_Y, α_X: how fast each series reacts to the gap (the adjustment coefficients).
- γ terms: short-run effects of last period's changes. They are separate from the long-run relation.
- Every term is either a difference (stationary) or the stationary z, so this regression has no spurious-regression problem.

3. The vector error-correction model (VECM)

A VECM is "a differenced VAR + memory of the equilibrium". Stacking the two-variable ECM of Section 2 in matrix form gives the same Johansen form seen in Part 2.

$$\Delta \mathbf{Y}_t = \alpha \underbrace{(\beta^\top \mathbf{Y}_{t-1})}_{z_{t-1}} + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t$$

Part	Role
ΣΓᵢΔY_{t−i}	An ordinary VAR on the changes. It captures short-run momentum and noise. Important for fit, but unrelated to the long-run tie
α(β^⊤Y_{t−1})	Last period's equilibrium error z_{t−1}. Exactly the term a differenced VAR leaves out. All of the long-run behaviour lives here

Differencing alone throws away the long-run relation. A VECM puts it back as an explicit correction term. That is why it is the natural model for cointegrated series. The only difference from the Week 6 VAR is this one term.

Adjustment speed α: sign, size, and who moves

α is the coefficient on last period's equilibrium error. It sets what percentage of the gap closes in one period.

- **Sign**: it must be negative to push the gap towards 0. A positive value widens the gap and explodes, so it is rejected.
- **Size**: a large |α| means fast, reliable reversion and an attractive trade. Close to 0, the spread barely comes back and capital is tied up for a long time.
- **Who moves**: each series has its own α. A series with α near 0 barely moves to close the gap, and the series with the larger |α| does most of the correcting. For example, in a pair where futures move first and spot follows, spot will show the larger |α|.

Combining the two equations into one for the spread z, the spread's own adjustment speed is a combination of the two α's.

$$\Delta z_t = \Delta Y_t - \beta\,\Delta X_t = \underbrace{(\alpha_Y - \beta\,\alpha_X)}_{\alpha_z\ <\ 0}\ z_{t-1} + \cdots$$

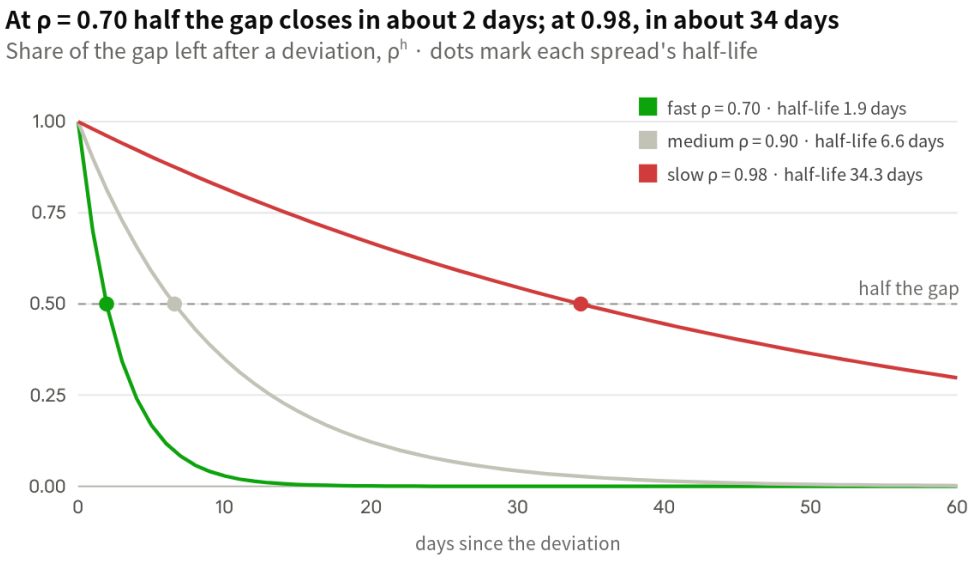
Half-life: turning α into days

Ignoring the short-run terms, the spread is AR(1). A share ρ = 1 + α_z of the gap remains each period.

$$z_t = (1+\alpha_z)\,z_{t-1} + u_t = \rho\,z_{t-1} + u_t, \qquad -1 < \alpha_z < 0 \iff 0 < \rho < 1$$

The gap left after h periods is ρ to the power h. The h at which it reaches one half is the half-life.

$$\rho^h = \tfrac{1}{2} \implies h_{1/2} = \frac{\ln 0.5}{\ln(1+\alpha_z)} = \frac{\ln 0.5}{\ln \rho}$$



Half-life formula computed for ρ = 0.70, 0.90, 0.98 from the lecture figure

- With ρ = 0.90 (α_z = −0.10), the half-life is about 6.6 days. That amounts to 10% of the gap closing each day.
- The half-life turns an abstract α into a concrete length of time. It sets a sensible holding period, screens out slow spreads not worth tying capital up in, and lets you check that the estimated speed is plausible.
- Comparing half-lives of 3 days and 60 days, the 3-day one is more tradable. Capital turns over more often and for shorter spells.

Granger representation theorem

$$\text{cointegrated} \iff \text{an error-correction representation exists} \quad \text{(Granger representation theorem)}$$

- If series are cointegrated, an error-correction representation exists; if an error-correction model holds, the series are cointegrated. Never one without the other.
- So modelling cointegrated series with a VECM is justified, and the long-run equilibrium is linked to the short-run dynamics. This is the theoretical backbone of the whole week.

4. Pairs trading Stage II: the cointegrated spread

If Week 5 gave pairs trading volatility-scaled bands, this week gives it a real anchor. The spread rests on an equilibrium, not on a fitted correlation.

What Stage II adds	The trade in one sentence
A statistically validated mean-reverting spread	Bet against the stretch, collect when it snaps back
Hedge ratio β from the cointegrating relation	Buy the cheap leg, sell the expensive leg —

What Stage II adds	The trade in one sentence
	market-neutral
A restoring force whose speed and half-life can be measured	Exit when the equilibrium error returns to 0

z-score signal

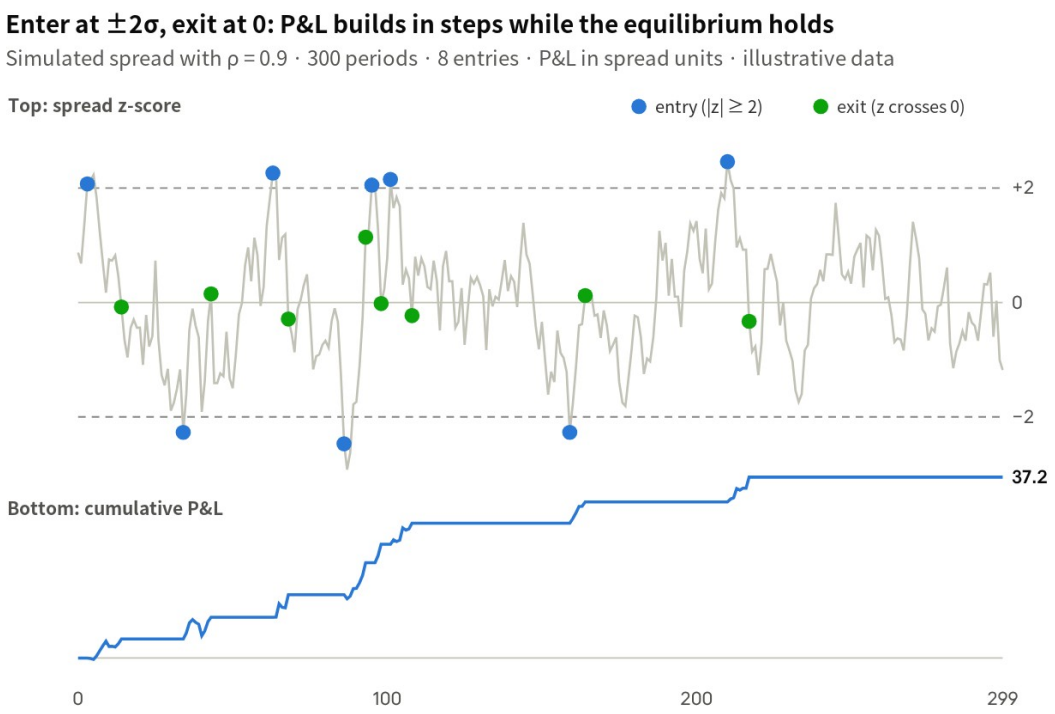
Standardise the spread by how many standard deviations it lies from its mean.

$$Z_t = \frac{z_t - \mu_z}{\sigma_z}, \qquad |Z_t| \geq 2 : \text{ enter against the stretch}, \qquad Z_t \to 0 : \text{ exit}$$

- Standardisation puts every pair on the same scale, so a threshold such as "2σ" becomes a universal entry rule.
- A large |Z| means the pair is unusually stretched. The signal is tied directly to the equilibrium error.
- In real trading, μ and σ must be estimated from a past (rolling) window only. Using the full sample leaks future information (look-ahead).

Entry and exit rules

	Rule
Entry	Z ≥ +2: Y is expensive → sell the spread (sell Y, buy β units of X). Z ≤ −2: the reverse, buy the spread
Exit	Exit when the spread returns towards equilibrium (Z = 0). Some profit may be taken earlier
Threshold	Higher thresholds give fewer but stronger signals
Stop-loss	Always set alongside, in case reversion fails (Section 5)



pairs.csv · AR(1) ρ = 0.9 simulation, 300 periods (illustrative data)

In the top panel, blue dots mark entries where |z| crosses 2, and green dots mark exits where z crosses 0. As long as the equilibrium holds, the cumulative P&L in the bottom panel steps up by one notch per trade.

Position sizing and market neutrality

$$\text{position} = \pm\,(1 \text{ unit of } Y,\ -\beta \text{ units of } X)$$

$$\text{P\&L}_t = \text{pos}_{t-1}\,\Delta z_t, \qquad \text{size} \propto \sigma_z^{-1}$$

- **Hedge ratio**: the cointegrating β is the quantity of X that hedges one unit of Y. Trading in that ratio makes the position market-neutral, so the bet is not on the direction of either market but on the **spread**.
- **Size**: divide by the spread's volatility (Week 5) to hit a target risk level. A wide, risky spread gets a smaller notional amount. Volatility (Week 5) and cointegration (Week 7) meet here.

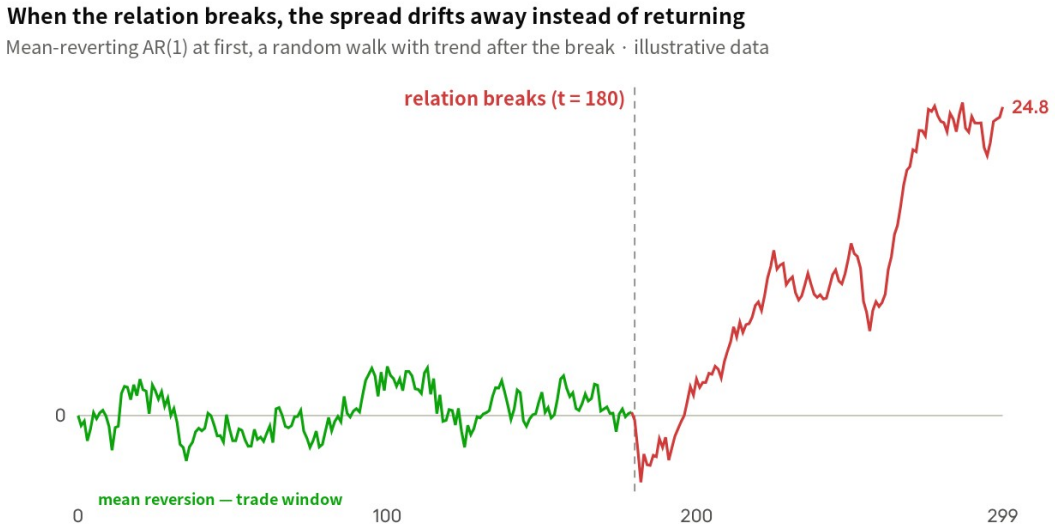
The full pairs-trading pipeline

1. **Find**: screen candidate pairs and check each is I(1) (Weeks 2 and 7).
2. **Test**: test for cointegration and estimate β (Week 7 Part 2).
3. **Spread**: form z = Y − βX and convert it to a z-score (Week 7 Part 3).
4. **Trade**: enter at |Z| ≥ 2, exit near 0, market-neutral.
5. **Manage**: stop-loss and regular re-testing for cointegration.

Three things act as the engine underneath. Cointegration provides the anchor (a stationary spread), error correction explains why it comes back, and volatility (Week 5) sets the bands and the size.

5. When cointegration breaks down

Even a relation that held for years can break. After a merger or acquisition, a regulatory change or a shift in business model, the spread no longer comes back but drifts away. The 'sure' trade turns into a loss.



breakdown.csv · 300 periods, structural break at t = 180 (illustrative data)

Why it happens	Risk
The economic link tying the pair changes or disappears	The spread widens instead of returning, and losses mount
Structural breaks, regime shifts, corporate events	"It always came back before" is the trap
The old β no longer describes the relation	The inherent risk of statistical arbitrage

Managing breakdown risk

- **Always keep a stop-loss.** It limits the loss when reversion fails. For example, exit if |Z| exceeds 4 or the holding period exceeds some multiple of the half-life.
- **Re-test cointegration regularly.** If it weakens, close the position. Re-estimate β as well.
- **Diversify across many pairs.** Size positions so that one broken pair is not fatal.
- **Mindset**: treat cointegration as a working assumption, not a law. Watch for economic events that could break the link (mergers, regulation, restructuring).

6. Week 7 wrap-up

Part	Key idea
Part 1 — Cointegration	When two I(1) series share one common trend, a stationary combination remains. It is the long-run equilibrium and separates real relations from spurious regression
Part 2 — Tests	Engle-Granger and Johansen prove the equilibrium. Johansen also reports how many equilibria there are (the rank)
Part 3 — Error correction and trading	Error correction is the restoring force, measured by its speed (α) and half-life. Pairs trading Stage II trades the z-score spread market-neutrally, and a breakdown of cointegration is the decisive risk

Next, Week 8 covers state-space models and the Kalman filter, plus the midterm exam.

7. Common misconceptions and self-check

Common misconception	Reality
A cointegrated pair always comes back	Cointegration can break down. Re-test, and never assume it lasts forever (Section 5)
The spread always comes back, so no stop-loss is needed	Always keep a stop-loss. A broken pair drifts away without limit (Section 5)
A past hedge ratio stays fixed forever	Both the relation and β can change. Monitor and re-estimate (Sections 4 and 5)

Three questions before wrapping up

1. **What does a negative α mean in a VECM?** The restoring force that pulls the equilibrium error back towards 0, that is, the adjustment speed.
2. **Which spread is more tradable, one with a half-life of 3 days or of 60 days?** The 3-day one. It comes back faster and ties up less capital.
3. **The spread drifts away instead of coming back. What has happened?** The cointegrating relation has broken down.